

CHAPTER 9

9.1) THERMAL EXPANSION NEAR ABSOLUTE ZERO

(a) From (7), (8) and (9):

$$\left(\frac{\partial V}{\partial \tau}\right)_{p,N} = \left(\frac{\partial}{\partial \tau} \frac{\partial G}{\partial p}\right)_N = \left(\frac{\partial}{\partial p} \frac{\partial G}{\partial \tau}\right)_N = - \left(\frac{\partial \sigma}{\partial p}\right)_{\tau,N} \quad (45a)$$

$$\left(\frac{\partial V}{\partial N}\right)_{p,\tau} = \left(\frac{\partial}{\partial N} \frac{\partial G}{\partial p}\right)_{\tau} = \left(\frac{\partial}{\partial p} \frac{\partial G}{\partial N}\right)_{\tau} = + \left(\frac{\partial \mu}{\partial p}\right)_{N,\tau} \quad (45b)$$

$$\left(\frac{\partial \mu}{\partial \tau}\right)_{N,p} = \left(\frac{\partial}{\partial \tau} \frac{\partial G}{\partial N}\right)_p = \left(\frac{\partial}{\partial N} \frac{\partial G}{\partial \tau}\right)_p = - \left(\frac{\partial \sigma}{\partial N}\right)_{\tau,p} \quad (45c)$$

(b) For physically reasonable systems, the ground state multiplicity g , and with it $\sigma = \log g$, go to constant limiting (small) values independent of pressure, hence $(\partial \sigma / \partial p)_{\tau} \rightarrow 0$. Because of (45a) this implies $(\partial V / \partial \tau)_p \rightarrow 0$ and hence $\alpha \rightarrow 0$.

9.2) THERMAL IONIZATION OF HYDROGEN

(a) The reaction equation $e + H^+ = H$ implies $v(e) = v(H^+) = 1$, $v(H) = -1$. If we neglect spin, neither the electron nor the proton have an internal free energy: $F(e) = F(H^+) = 0$. Hence, from (34):

$$K(\tau) = [n_Q(e)n_Q(H^+)/n_Q(H)] \exp[+F(H)/\tau] \quad .$$

Because the mass of the proton and the mass of the hydrogen atom are very nearly the same, the two terms $n_Q(H^+)$ and $n_Q(H)$ cancel out. If I is the energy required to ionize atomic hydrogen, then the internal free energy of the hydrogen atom relative to its separate constituents is $F(H) = -I$, in the absence of additional internal degrees of freedom (spin). Hence,

$$K(\tau) \cong n_Q(e) \exp(-I/\tau) \quad ,$$

which is equivalent to (47).

If we had included the electron spin, it would have contributed a factor $Z_{\text{int}} = 2$ to the partition function and a term $\tau \log 2$ to the internal free energy of both the electron and the hydrogen atom. These terms would have cancelled out in (34) because $v(e) = -v(H)$. The effects of the proton spin cancel out similarly.

(b) The first excited state of the hydrogen atom is four-fold degenerate, not counting spin. Hence, from (3.9):

$$[H(\text{exc})]/[H(\text{grd})] = P(\text{exc})/P(\text{grd}) = 4 \exp(-3I/4\tau) \quad .$$

Numerically $I = 1$ Rydberg = 13.61 eV. With this and $[H] = 10^{23} \text{cm}^{-3}$, for $T = 5000$ K:

$$[H(\text{exc})] = 4 \times 10^{23} \text{cm}^{-3} \times \exp(-23.68) = 2.07 \times 10^{13} \text{cm}^{-3}.$$

To compare this with $[e]$, taken from (48), we note that at $T = 5000$ K we have $n_0 = 2.415 \times 10^{15} \text{cm}^{-3} \times (5000)^{3/2} \cong 8.54 \times 10^{20} \text{cm}^{-3}$. If this is inserted into (48): $[e] = (1 \times 10^{23} \times 8.54 \times 10^{20})^{1/2} \text{cm}^{-3} \times \exp(-15.79) \cong 1.28 \times 10^{15} \text{cm}^{-3}$. We see that the concentrations of both ionized and excited hydrogen atoms are small compared to those of the ground state hydrogen and hence of the total hydrogen. Note also that the concentration of ionized hydrogen is significantly larger than that of excited hydrogen, even though more energy is required to ionize hydrogen than to excite it. This is because the number of states accessible to separate electrons and protons is very much larger than those accessible to a hydrogen-proton pair bound to each other.

9.3) IONIZATION OF DONOR IMPURITIES IN SEMICONDUCTORS

If we treat both the free electrons and the donors as ideal gases, the problem of impurity ionization becomes formally identical to the preceding problem of the ionization of atomic hydrogen, leading to a relation of the form

(47). We pick up the trail at this point, re-writing (47) in the form

$$[e][n_d^+]/[n_d^0] = n_Q(m^*) \exp(-I^*/\tau) \equiv K \quad (S1)$$

Here $[n_d^+]$ and $[n_d^0]$ are the concentrations of ionized and un-ionized donors, $n_Q(m^*)$ is the quantum concentration for particles with mass m^* , and I^* the donor ionization energy. Note that the quantum concentrations for the donors cancel out, which signals that the equilibrium is independent of the mass of the donors, which means that it is independent of their motion. This is why we get away with treating the donors as an ideal gas.

To estimate I^* we recall from atomic physics that the ionization energy for the hydrogen atom is

$$(CGS) \quad I = \frac{e^4 m}{2\hbar^2} \quad , \quad (SI) \quad I = \frac{e^4 m}{2(4\pi\epsilon_0\hbar)^2} \quad ,$$

where the symbol ϵ_0 in the SI expression denotes the permittivity of space, not an energy. Numerically, $I = 1$ Rydberg = 13.6 eV. In a semiconductor, we must replace ϵ_0 by $\epsilon\epsilon_0$ (or e^2 by e^2/ϵ) and m by m^* . With $\epsilon = 11.7$, $T = 100$ K, $m^* = 0.3m$ we obtain

$$I^* = I \times (m^*/m)/\epsilon^2 \cong 0.0298 \text{ eV} \quad , \quad I/\tau = 3.46 \quad .$$

Similarly, in n_Q , m must be replaced by m^* :

$$\begin{aligned} n_Q(m^*) &= (2.415 \times 10^{15} \text{ cm}^{-3}) \times [(m^*/m) \times (T/1K)]^{3/2} \\ &= 3.97 \times 10^{17} \text{ cm}^{-3} \quad . \end{aligned}$$

With these values:

$$K = 1.25 \times 10^{16} \text{ cm}^{-3} \quad .$$

This value is sufficiently close to the total donor concentration $[n_d] = 10^{17} \text{ cm}^{-3}$ that a substantial fraction of the donors will be ionized, and we cannot simply set

$[n_d^0] \cong [n_d]$. Instead, we must set $[n_d^0] = [n_d] - [n_d^+]$
 $= [n_d] - [e]$. If this is inserted into (S1), a quadratic
equation for $[e]$ results, the positive root of which is

$$[e] = [(K/2)^2 + K n_d]^{1/2} - K/2 \cong 2.97 \times 10^{16} \text{cm}^{-3} ,$$

corresponding to about 30% ionization.

A more rigorous treatment of impurity ionization in semi-conductors, leading to essentially the same result without treating the impurities as ideal gas atoms, is given in Chapter 13.

9.4) BIOPOLYMER GROWTH

(a) We have, for the basic reaction steps,

$$\left. \begin{aligned} [1][1]/[2] &= K_1, [1][2]/[3] = K_2, \dots, \\ [1][N]/[N+1] &= K_N. \end{aligned} \right\} \quad (S1)$$

Multiplication of all equations yields

$$[1]^{N+1}/[N+1] = K_1 K_2 \dots K_N ,$$

which is equivalent to (49).

(b) Eq. (50) follows directly by inserting $v(1) = v(N) = -v(N+1) = 1$ into (34).

(c) When $n_Q(N+1) \cong n_Q(N)$, (50) simplifies to

$$K_N \cong n_Q(1) \exp(\Delta F/\tau) . \quad (S2)$$

Then, for $\Delta F = 0$, from (S1):

$$[N+1]/[N] = [1]/K_N \cong [1]/n_Q(1) .$$

Numerically, for $M = 100 \text{ amu}$, $T = 300 \text{ K}$:

$$\begin{aligned} n_Q(1) &= (1.879 \times 10^{20} \text{cm}^{-3}) \times [(M/1 \text{amu}) \times (T/1 \text{K})]^{3/2} \\ &= 2.76 \times 10^{27} \text{cm}^{-3} . \end{aligned}$$

Hence, with $[1] = 10^{20} \text{cm}^{-3}$ we have $[N+1]/[N] \cong 3.6 \times 10^{-8}$.

(d) To obtain large polymers, we need $[N+1] \geq [N]$, which requires, from (S1), $K_N \leq [1]$. Inserted into (S2):

$$\Delta F \leq -\tau \log\{n_Q(1)/[1]\} \cong -0.44 \text{ eV} .$$

9.5) PARTICLE-ANTIPARTICLE EQUILIBRIUM

(a) We have $v_+ = v_- = 1$, $[A^+] = [A^-] = n$. If the zero of the energy scale is the energy with no particles present, then zero kinetic energy of the particles corresponds to a finite "internal" energy, such that $F_+(\text{int}) + F_-(\text{int}) = \Delta$, the energy required to create a particle-antiparticle pair at rest. If we insert these values into (34), the mass action law (35) assumes the form

$$[A^+][A^-] = n_+ n_- = n^2 = n_Q^2 \exp[-\Delta/\tau] ;$$

$$n = n_Q \exp[-\Delta/2\tau] .$$

(b) For electrons at 300 K we have $n_Q = 1.25 \times 10^{19} \text{cm}^{-3}$; hence, with $\Delta/2\tau = 10$: $n = 5.7 \times 10^{14} \text{cm}^{-3}$.

(c) For particles with spin $\frac{1}{2}$, n_Q is replaced by $2n_Q$ -- see (6.47) --, hence

$$n = 2n_Q \exp[-\Delta/2\tau] .$$

Numerically, for the case under (b): $n = 1.14 \times 10^{15} \text{cm}^{-3}$.